



AI Classical Systems: General Problem Solver (GPS)

1. Background and Context

- In the early days of Artificial Intelligence (1950s-60s), researchers aimed to build machines that could mimic **human problem-solving**.
 - Early AI was based on **symbolic processing**, where problems were represented by symbols and rules.
 - Allen Newell and Herbert A. Simon, pioneers in AI, developed the **General Problem Solver (GPS)** in 1957 as a universal problem-solving program.
 - GPS was inspired by human **means-ends problem solving** methods observed in psychology.
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2. What is the General Problem Solver?

- GPS is a **computer program** designed to solve problems by searching through a **problem space** of possible states.
 - It tries to find a **sequence of operators (actions)** to move from an initial state to a goal state.
 - It is **domain-independent**, meaning it can theoretically solve any problem, provided it can be represented correctly.
 - GPS uses a **heuristic search strategy** known as **means-ends analysis** to guide problem solving.
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3. Core Concepts in GPS

a) Problem Representation

- **Initial State (S₀):** The starting point of the problem.
- **Goal State (G):** The desired end state.
- **Operators (O):** Defined actions that transform one state into another.
- **Problem Space:** The entire network or graph of possible states and transitions formed by applying operators.

Example:

In the Tower of Hanoi puzzle, states represent configurations of disks on pegs, operators are moves that transfer a disk from one peg to another, initial state is all disks stacked on one peg, and goal state is disks stacked on another peg in the same order.

b) States and Operators Formalization

- States are often represented symbolically, for example:
 - **State S:** A list or structure describing the current configuration.
- Operators have:
 - **Preconditions:** Conditions that must hold true to apply the operator.
 - **Effects:** How the state changes after the operator is applied.

Example:

Operator `Move(Disk1, PegA, PegB)`

Precondition: Disk1 is on PegA, and PegB's top disk is larger or PegB is empty.

Effect: Disk1 moves to PegB.



4. Means-Ends Analysis: The Heart of GPS

- GPS attempts to reduce the difference between the **current state** and the **goal state**.
 - The approach breaks the overall problem into **subproblems** by focusing on the largest or most significant difference first.
 - For each difference, GPS:
 1. Identifies an operator that can reduce that difference.
 2. Attempts to apply the operator.
 3. If applying the operator is blocked, GPS treats the precondition of the operator as a new subgoal to solve.
 - This is a **recursive process**, continually breaking down goals until reaching easily solvable subgoals.
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Example of Means-Ends Analysis:

- Goal: Get disk from Peg A to Peg C.
- Current state: Disk on Peg A.
- Difference: Disk not on Peg C.
- Operator: Move disk from Peg A to Peg C.
- Precondition: Peg C must be free or have larger disk on top.
- If precondition not met, set new subgoal: Clear Peg C.
- Solve subgoal first (e.g., move disk from Peg C to Peg B).
- Once subgoal solved, apply original operator.



5. GPS Process Flow and Algorithm

1. **Initialize** with initial state (S_0) and goal state (G).
2. **Difference Identification:** Compare current state (S) to goal (G) to find differences.
3. **Operator Selection:** Choose an operator O that can reduce the identified difference.
4. **Operator Application:**
 - Check if preconditions of O are met.
 - If yes, apply $O \rightarrow$ new state S' .
 - If no, treat unmet preconditions as new subgoals.
5. **Recursive Subgoal Solving:** Solve each subgoal using the same means-ends process.
6. **Backtracking:**
 - If operator application or subgoal solving fails, backtrack to try alternate operators.
7. **Termination:** When current state equals goal state, return the sequence of operators as solution.

6. Formal Representation of GPS Search

- GPS searches a **tree/graph** of states:
 - Nodes = states.
 - Edges = operators.



- Search strategy is essentially **depth-first with backtracking**, driven by heuristic means-ends analysis rather than blind exploration.

7. GPS Rules Explained in Depth

Rule	Explanation	Example/Implication
1. Define Problem Precisely	GPS needs clear formal definitions of initial state, goal state, and operators (actions). Without this, GPS cannot function.	Without operator definitions, GPS cannot apply changes.
2. Identify Differences	GPS constantly measures difference between current and goal state to guide operator selection.	If goal is "disk on peg C," difference is disk not on peg C.
3. Select Operators That Reduce Differences	Operators are chosen only if they help reduce the difference; irrelevant operators are discarded.	Moving unrelated disks won't be selected if they don't help move target disk.
4. Check Preconditions	Before applying an operator, GPS verifies if preconditions are met. If not, preconditions become subgoals to solve.	Move disk only if target peg is free or has larger disk.
5. Recursively Solve Subgoals	If an operator's preconditions are not met, GPS recursively solves for those preconditions as new goals.	Clearing a peg before moving disk there.
6. Backtracking	If operator or subgoal fails, GPS backtracks to try other operators or paths.	If moving disk to peg B fails, try moving disk to peg C.
7. Terminate When Goal Reached	Process ends when current state matches the goal state.	Puzzle solved!



8. Advantages of GPS

- **Domain independence:** The same GPS program can solve many different problems just by changing the problem definition.
 - **Heuristic approach:** Means-ends analysis efficiently narrows the search space.
 - **Demonstrated feasibility:** Proved that problem solving could be formalized and automated.
 - **Foundation for AI research:** Paved the way for heuristic search algorithms, planning, and automated reasoning.
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9. Limitations and Criticisms

Limitation	Explanation	Example
Requires Precise Formalization	The problem and operators must be completely and precisely defined.	Real-world problems often lack clear definitions.
Inefficiency for Large Spaces	Depth-first search with backtracking can be very slow or get stuck in large or complex spaces.	GPS may take exponential time for large puzzles.
No Learning or Adaptation	GPS does not learn from previous attempts or adjust its strategy dynamically.	GPS repeats the same steps if restarted.
Not Good with Ambiguous or Incomplete Info	Cannot handle uncertainty or partial knowledge.	Cannot work well in real-world dynamic environments.
Limited to Symbolic Reasoning	Does not handle numeric computation or probabilistic reasoning well.	Cannot solve problems needing statistical inference.

10. Modern Implications and Legacy

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- GPS introduced the **means-ends analysis heuristic**, still fundamental in AI planning.
 - Modern AI systems use more sophisticated search methods: *A search, constraint satisfaction, reinforcement learning**.
 - GPS concepts influence **automated planning systems** and **rule-based expert systems**.
 - However, symbolic AI alone is no longer enough; it is combined with **machine learning** and **statistical methods** in today's AI.
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11. Summary & Key Takeaways

- GPS is a pioneering symbolic AI system for universal problem solving.
 - It models problem solving as a **search through problem space** using **operators**.
 - **Means-ends analysis** guides operator selection by reducing differences.
 - GPS operates recursively, handling subgoals via operator preconditions.
 - The system uses depth-first search with backtracking.
 - GPS is powerful conceptually but limited by efficiency and real-world applicability.
 - It laid the groundwork for many modern AI problem-solving and planning techniques.
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12. Additional Illustrative Example: The Missionaries and Cannibals Problem

- Problem: 3 missionaries and 3 cannibals must cross a river with a boat that can hold 2 people.



- Constraints: Cannibals cannot outnumber missionaries on either side.
- GPS states: Configuration of missionaries, cannibals, and boat position.
- Operators: Move one or two people across the river, checking constraints.
- GPS uses means-ends to reduce difference (people on the wrong side) while respecting constraints.
- Backtracking needed when constraints are violated.